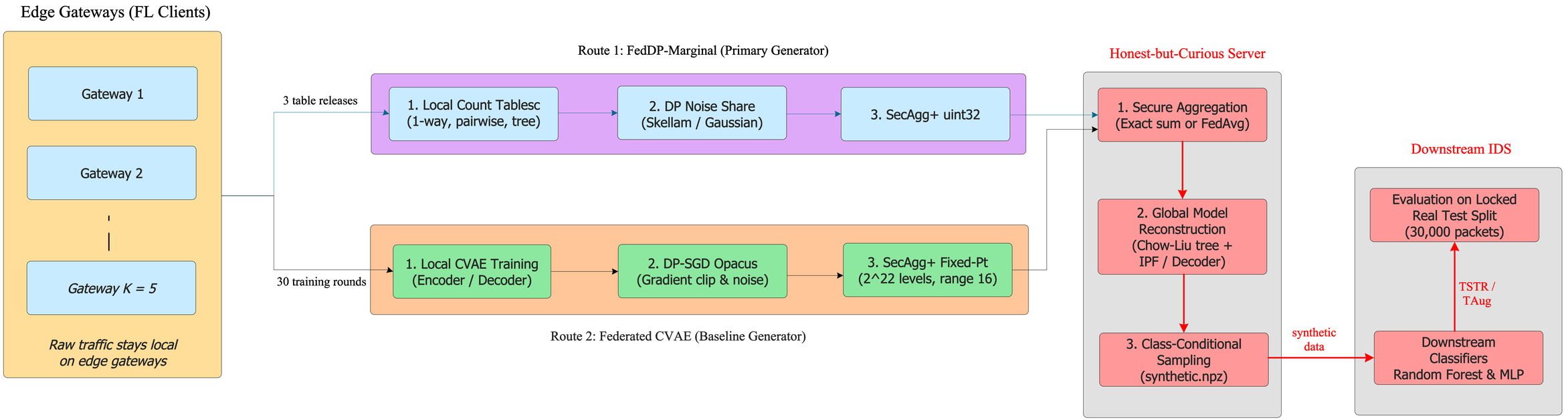

Edge Gateways (FL Clients)
Gateway 1
Gateway 2
Gateway K = 5
Raw traffic stays local on edge gateways
Route 1: FedDP-Marginal (Primary Generator)
3 table releases
1. Local Count Tablesc (1-way, pairwise, tree)
2. DP Noise Share (Skellam / Gaussian)
3. SecAgg+ uint32
30 training rounds
1. Local CVAE Training (Encoder / Decoder)
2. DP-SGD Opacus (Gradient clip & noise)
3. SecAgg+ Fixed-Pt (2^22 levels, range 16)
Route 2: Federated CVAE (Baseline Generator)
Honest-but-Curious Server
1. Secure Aggregation (Exact sum or FedAvg)
2. Global Model Reconstruction (Chow-Liu tree + IPF / Decoder)
3. Class-Conditional Sampling (synthetic.npz)
synthetic data
Downstream IDS
Evaluation on Locked Real Test Split (30,000 packets)
TSTR / TAug
Downstream Classifiers Random Forest & MLP